Topic 1: Simple Linear Regression Chapter 12 of our Textbook

STAT 2509: Introduction to Statistical Modeling II

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January 23, 2026

Table of Contents

1. Introduction and Motivation
2. Model Assumptions (LINE)
3. Parameter Estimation
4. Statistical Inference
5. Prediction
6. Analysis of Variance (ANOVA) and F-test
7. Model Assessment
8. Lack of Fit Test
9. Simple Linear Regression (SLR) Formulae

What is a Probabilistic Model?

Deterministic Model:

$$y = \beta_0 + \beta_1 x$$

- Exact relationship
- No randomness
- Example: $F = 32 + 1.8C$

Probabilistic Model:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- Includes randomness
- ε = random error
- Real-world data: income vs education, drug dose vs response

Example: Study Hours vs Exam Score

Data from 10 students:

Hours (x)	2	3	4	5	6	7	5	5	1	5
Score (y)	65	70	75	78	80	85	80	78	58	88

Probabilistic Model:

$$\text{Score}_i = \beta_0 + \beta_1(\text{Hours}_i) + \varepsilon_i$$

- ε_i represents: sleep quality, prior knowledge, test anxiety, etc.
- Not all students who study 5 hours get the same score.

Simple Linear Regression (SLR)

Let y and x be two variables observed on experimental units $i = 1, 2, \cdots, n$. **A Simple Linear Regression** model assumes that:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \text{for all } i,$$

where

- β_0 & β_1 are two unknown parameters (regression coefficients) called **intercept** & **slope**, respectively.
- ε is the error term (random error).
- x: called **independent** variable, or **predictor** or **explanatory**.
- y: called **dependent** variable, or **response** or **covariate**.

The LINE Assumptions

1. **L**inearity: The expected value (average) of Y given $X = x$ is linear in the parameters β_0 and β_1

$$E(y_i|x_i) = \beta_0 + \beta_1 x_i$$

2. **I**ndependence: Errors $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are independent
3. **N**ormality:

$$\varepsilon_i \sim N(\text{mean } = 0, \text{variance } = \sigma^2)$$

4. **E**qual variance (Homoscedasticity):

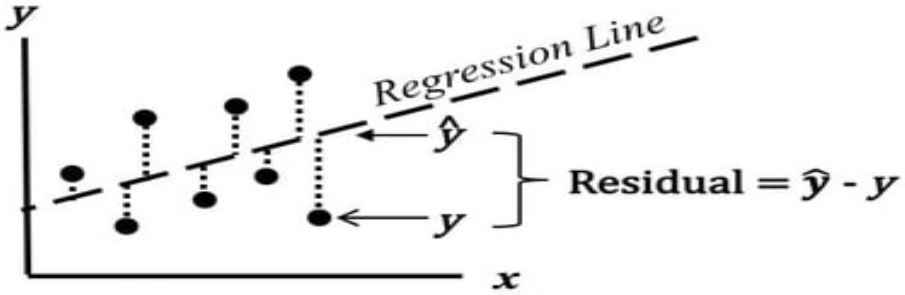
$$\text{Var}(\varepsilon_i) = \sigma^2 \quad \text{for all } i$$

5. x_i's are observed without errors.

Least Squares Estimation (LSE)

Goal: Minimize Sum of Squared Errors (SSE)

$$SSE = \sum_{i=1}^{n} [y_i - \hat{y}_i]^2 = \sum_{i=1}^{n} [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)]$$



Sample (or Estimated or Predicted) Regression Line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \qquad \text{and} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2 = \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} = \sum_{i=1}^{n} x_i y_i - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i}{n}$$

Note: $\hat{\varepsilon}_i = e_i = y_i - \hat{y}_i$ is called **residual** and $\sum_{i=1}^{n} e_i = 0$.

Example

10 students data:

Hours (x)	2	3	4	5	6	7	5	5	1	5
Score (y)	65	70	75	78	80	85	80	78	58	88

Given: $n = 10, \sum x = 43, \sum y = 757, \sum xy = 3393, \sum x^2 = 215$

1. Means:

$$\bar{x} = \frac{\sum x}{n} = \frac{43}{10} = 4.3, \quad \bar{y} = \frac{\sum y}{n} = \frac{757}{10} = 75.7$$

2. Regression coefficients:

$$\hat{\beta}_1 = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} = \frac{3393 - \frac{43 \cdot 757}{10}}{215 - \frac{43^2}{10}} = \frac{137.9}{30.1} \approx 4.5814$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 75.7 - 4.581 \times 4.3 \approx 56.002$$

3. Regression equation:

$$\hat{y} = 56.002 + 4.581x$$

4. Prediction for $x = 4$:

$$\hat{y} = 56.002 + 4.581 \times 4 \approx 74.326$$

5. Error (actual $y = 75$):

$$\text{Error} = y_{x=4} - \hat{y}_{x=4} = 75 - 74.326 \approx 0.674$$

Interpretation of Regression Parameters

Interpretation Template

- **Intercept:** When $x = 0$, we predict y to be $\hat{\beta}_0$ units. This represents the baseline level of the response when the predictor is absent.
- **Slope:** For every additional unit of x, we expect y to change by $\hat{\beta}_1$ units. A positive slope indicates increase, negative indicates decrease.

10 students data

- **Intercept:** Students with zero study hours are predicted to score approximately 56 points on the exam.
- **Slope:** Each additional hour of studying increases the predicted exam score by about 4.58 points.

Estimating $\text{Var}(\varepsilon) = \sigma^2$

$$S^2 = \frac{SSE}{n-2} = \frac{\sum_{i=1}^{n} e_i^2}{n-2} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n-2} = MSE = \text{Mean Squares Errors}$$

Calculations for each observation (10 students data):

x_i	y_i	$\hat{y}_i = 56.002 + 4.581x_i$	$e_i = y_i - \hat{y}_i$	e_i^2
2	65	65.164	-0.164	0.027
3	70	69.745	0.255	0.065
4	75	74.326	0.674	0.454
5	78	78.907	-0.907	0.823
6	80	83.488	-3.488	12.166
7	85	88.069	-3.069	9.419
5	80	78.907	1.093	1.195
5	78	78.907	-0.907	0.823
1	58	60.583	-2.583	6.672
5	88	78.907	9.093	82.683
Total				**114.327**

$$S^2 = \frac{SSE}{n-2} = \frac{114.327}{8} \approx 14.29$$

Inference for Slope and Intercept

Sampling Distributions

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$
$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]\right)$$

Standard Errors

$$\text{SE}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}$$
$$\text{SE}(\hat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

where $s = \sqrt{s^2} = \sqrt{\frac{SSE}{n-2}}$ is called the **Residual Standard Error**.

Inference for Slope and Intercept

Hypothesis Test for Slope

Test $H_0 : \beta_1 = 0$ vs $H_a : \beta_1 \neq 0$ (no linear relationship vs some linear relationship).
Test statistic:

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}\left(\hat{\beta}_1\right)} \sim t_{n-2} \quad \text{under } H_0$$

Reject H_0 if $|t| > t_{\alpha/2,n-2}$ or if p-value $< \alpha$.

Hypothesis Test for Intercept

Test $H_0 : \beta_0 = 0$ vs $H_a : \beta_0 \neq 0$ (Regression line passes through the origin vs not).
Test statistic:

$$t = \frac{\hat{\beta}_0 - 0}{\text{SE}\left(\hat{\beta}_0\right)} \sim t_{n-2} \quad \text{under } H_0$$

Reject H_0 if $|t| > t_{\alpha/2,n-2}$ or if p-value $< \alpha$.

Inference for Slope and Intercept

Confidence Interval for Slope

$$\hat{\beta}_1 \pm t_{\alpha/2,n-2} \cdot \mathrm{SE}\left(\hat{\beta}_1\right) \equiv \left(\hat{\beta}_1 - t_{\alpha/2,n-2} \cdot \frac{s}{\sqrt{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2,n-2} \cdot \frac{s}{\sqrt{S_{xx}}}\right)$$

The interpretation of a 95% confidence interval is as follows:
"We are 95% confident that the true mean change in the dependent variable (Y) for every one-unit increase in the independent variable (X) falls somewhere within this interval."

Confidence Interval for Intercept

$$\hat{\beta}_0 \pm t_{\alpha/2,n-2} \cdot \mathrm{SE}\left(\hat{\beta}_0\right) \equiv \left(\hat{\beta}_0 - t_{\alpha/2,n-2} \cdot s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}, \hat{\beta}_0 - t_{\alpha/2,n-2} \cdot s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}\right)$$

The interpretation of a 95% confidence interval is as follows:
"We are 95% confident that the true mean value of the dependent variable (Y) when the independent variable (X) is equal to zero falls somewhere within this interval."

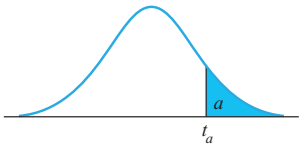


TABLE 4
Critical Values of *t*

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	*df*
1	3.078	6.314	12.706	31.821	63.657	1
2	1.886	2.920	4.303	6.965	9.925	2
3	1.638	2.353	3.182	4.541	5.841	3
4	1.533	2.132	2.776	3.747	4.604	4
5	1.476	2.015	2.571	3.365	4.032	5
6	1.440	1.943	2.447	3.143	3.707	6
7	1.415	1.895	2.365	2.998	3.499	7
8	1.397	1.860	2.306	2.896	3.355	8
9	1.383	1.833	2.262	2.821	3.250	9
10	1.372	1.812	2.228	2.764	3.169	10
11	1.363	1.796	2.201	2.718	3.106	11
12	1.356	1.782	2.179	2.681	3.055	12
13	1.350	1.771	2.160	2.650	3.012	13
14	1.345	1.761	2.145	2.624	2.977	14
15	1.341	1.753	2.131	2.602	2.947	15
16	1.337	1.746	2.120	2.583	2.921	16
17	1.333	1.740	2.110	2.567	2.898	17
18	1.330	1.734	2.101	2.552	2.878	18
19	1.328	1.729	2.093	2.539	2.861	19
20	1.325	1.725	2.086	2.528	2.845	20
21	1.323	1.721	2.080	2.518	2.831	21
22	1.321	1.717	2.074	2.508	2.819	22
23	1.319	1.714	2.069	2.500	2.807	23
24	1.318	1.711	2.064	2.492	2.797	24
25	1.316	1.708	2.060	2.485	2.787	25
26	1.315	1.706	2.056	2.479	2.779	26
27	1.314	1.703	2.052	2.473	2.771	27
28	1.313	1.701	2.048	2.467	2.763	28
29	1.311	1.699	2.045	2.462	2.756	29
∞	1.282	1.645	1.960	2.326	2.576	∞

Source: From "Table of Percentage Points of the *t*-Distribution," *Biometrika* 32 (1941):300. Reproduced by permission of the *Biometrika* Trustees.

Error variance estimate (10 students data)

$$s^2 = MSE = \frac{SSE}{n-2} = \frac{114.327}{8} = 14.291$$
$$s = \sqrt{14.291} = 3.780$$

Interpretation: The typical deviation of actual scores from the regression line is about 3.78 points.

Standard errors:

$$\text{SE}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{3.780}{\sqrt{30.1}} = \frac{3.780}{5.486} = 0.689$$
$$\text{SE}(\hat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$
$$= 3.780\sqrt{\frac{1}{10} + \frac{(4.3)^2}{30.1}}$$
$$= 3.780\sqrt{0.1 + 0.614} = 3.780\sqrt{0.714} = 3.780 \times 0.845 = 3.195$$

Hypothesis test for slope (10 students data)

Test: $H_0 : \beta_1 = 0$ vs $H_a : \beta_1 \neq 0$

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} = \frac{4.581}{0.689} = 6.65$$

$$t_{0.025,8} = 2.306 \quad \text{(from t-table)}$$

$$6.65 > 2.306 \Rightarrow \text{Reject } H_0$$

95% Confidence Interval for β_1:

$$\hat{\beta}_1 \pm t_{0.025,8} \times \text{SE}(\hat{\beta}_1)$$

$$4.581 \pm 2.306 \times 0.689$$

$$4.581 \pm 1.589$$

$$= (2.992, 6.170)$$

Interpretation: We are 95% confident that each additional hour of studying increases exam scores by 2.99 to 6.17 points.

Inference for intercept (10 students data)

95% Confidence Interval for β_0:

$$\begin{aligned} \hat{\beta}_0 \pm t_{0.025,8} \times \text{SE}(\hat{\beta}_0) \\ 56.002 \pm 2.306 \times 3.195 \\ 56.002 \pm 7.367 \\ = (48.635, 63.369) \end{aligned}$$

Interpretation: We are 95% confident that a student who studies 0 hours would score between 48.6 and 63.4 points.

Caution: This is extrapolation since our data starts at 1 hour.

Two Types of Prediction

Mean Response

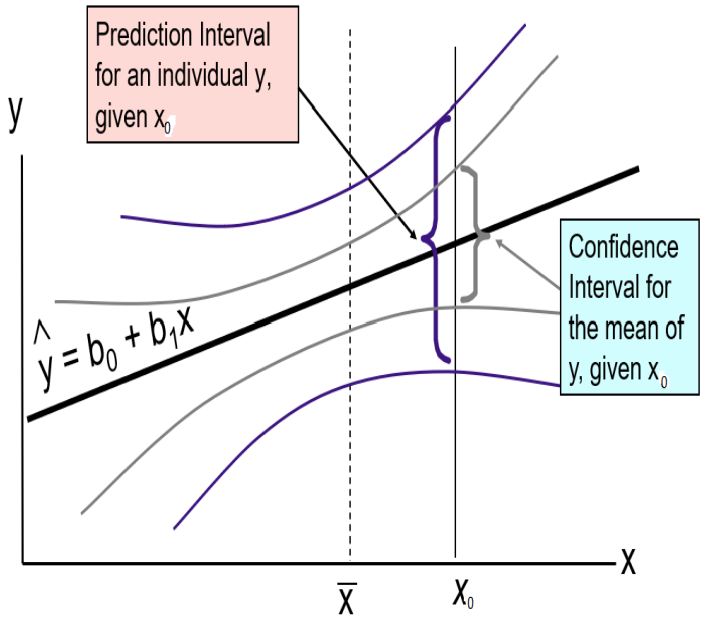
- Predict $E(y \mid x^\star)$.
- Average of all responses at $x^\star$.
- **Confidence Interval (CI)** for $E(y \mid x^\star)$:

$$\hat{y}_0 \pm t_{\alpha/2,n-2} \cdot s\sqrt{\frac{1}{n} + \frac{(x^\star - \bar{x})^2}{S_{xx}}}.$$

Individual Observation

- Predict single y at $x^\star$.
- Includes individual variation.
- **Prediction Interval (PI)** for y at $x^\star$:

$$\hat{y}_0 \pm t_{\alpha/2,n-2} \cdot s\sqrt{1 + \frac{1}{n} + \frac{(x^\star - \bar{x})^2}{S_{xx}}}.$$



Note: $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x^\star$ and PI is wider than Cl for mean response due to extra σ^2 term for individual variation.

Prediction for New Values (10 students data)

Predict for $x^\star = 4.5$ hours (average student)

$$\hat{y} = 56.002 + 4.581 \times 4.5 = 56.002 + 20.615 = 76.617$$

$$\begin{aligned}
\text{SE}_{\text{mean}} &= s\sqrt{\frac{1}{n} + \frac{(x^\star - \bar{x})^2}{S_{xx}}} \\
&= 3.780\sqrt{\frac{1}{10} + \frac{(4.5 - 4.3)^2}{30.1}} \\
&= 3.780 \times 0.318 = 1.202
\end{aligned}$$

95% CI for mean score at $x = 4.5$:

$$\begin{aligned}
&76.617 \pm 2.306 \times 1.202 \\
&= 76.617 \pm 2.771 = (73.846, 79.388)
\end{aligned}$$

Prediction Interval vs Confidence Interval (10 students data)

Predict for $x^\star = 5$ hours (with repeated data)

Point estimate:

$$\hat{y} = 56.002 + 4.581 \times 5 = 78.907$$

95% CI for mean score at $x = 5$:

$$\begin{aligned}
\text{SE}_{\text{mean}} &= 3.780\sqrt{\frac{1}{10} + \frac{(5-4.3)^2}{30.1}} \\
&= 3.780\sqrt{0.1 + \frac{0.49}{30.1}} \\
&= 3.780\sqrt{0.1 + 0.01628} = 3.780\sqrt{0.11628} = 1.287
\end{aligned}$$

$$\begin{aligned}
\text{CI: } & 78.907 \pm 2.306 \times 1.287 \\
& = 78.907 \pm 2.968 = (75.939, 81.875)
\end{aligned}$$

Actual data at $x = 5$: $78, 80, 78, 88$ (mean $= 81$, within CI)

Prediction Interval for Individual (10 students data)

$$\begin{aligned}\text{SE}_{\text{ind}} &= s\sqrt{1+\frac{1}{n}+\frac{(x^\star-\bar{x})^2}{S_{xx}}}\\ &= 3.780\sqrt{1+0.1+\frac{(5-4.3)^2}{30.1}}\\ &= 3.780\times 1.057 = 3.996\end{aligned}$$

95% PI for individual student at $x = 5$:

$$\begin{aligned}\text{PI: } &78.907 \pm 2.306 \times 3.996\\ &= 78.907 \pm 9.217 = (69.690, 88.124)\end{aligned}$$

Comparison:

- CI for mean: $(75.94, 81.88)$ - narrower
- PI for individual: $(69.69, 88.12)$ - wider
- All 4 actual scores at $x = 5$ fall within PI: $78, 80, 78, 88$

Extrapolation Warning (10 students data)

Predict for $x = 10$ hours (extrapolation!)

$$\hat{y} = 56.002 + 4.581 \times 10 = 101.812$$

Problem:

- Our data only goes to 7 hours
- The relationship might not be linear beyond 7 hours
- Diminishing returns likely after certain point
- Fatigue might reduce effectiveness

Warning: Predictions outside the data range ($x = 1$ to 7) are unreliable!

The analysis of variance (ANOVA)

Partitioning of Variation

- Total sum of squares *(Total variation)* is given by

$$TSS = \sum (y_i - \bar{y})^2 = S_{yy} = \sum y_i^2 - n\bar{y}^2 = \sum y_i^2 - \frac{(\sum y_i)^2}{n}.$$

- Sum of squares for regression *(Explained variation)* is given by

$$SSR = \sum (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}.$$

- Sum of squared errors *(Unexplained variation)* is given by

$$SSE = \sum (y_i - \hat{y}_i)^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}.$$

- Total variation is the sum of explained and unexplained variation. That is

$$TSS = SSR + SSE.$$

Degrees of Freedom:

- df(SSR) $= 1$
- df(SSE) $= n - 2$
- df(TSS) $= n - 1$

Partitioning of Variation Calculations

Total variation:

$$TSS = S_{yy} = 746.1 \quad (\text{df} = n - 1 = 9)$$

Regression variation:

$$SSR = \hat{\beta}_1^2 S_{xx} = (4.581)^2 \times 30.1$$

$$= 20.98 \times 30.1 = 631.773$$

Error variation:

$$SSE = 114.327 \quad (\text{df} = n - 2 = 8)$$

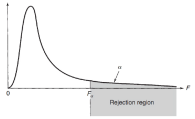
Check:

$$TSS = SSR + SSE$$

$$746.1 = 631.773 + 114.327 \quad \checkmark$$

Analysis of Variance (ANOVA) Table for Simple Linear Regression

Table: The analysis of variance (ANOVA)

Source	SS	df	MS	F	Rejection region
Regression	SSR	1	$MSR = \frac{SSR}{1}$	$\frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)}$	
Errors	SSE	$n-2$	$MSE = \frac{SSE}{n-2}$		
Total	TSS	$n-1$			

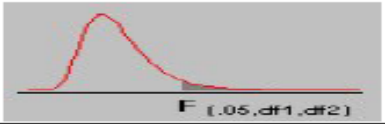
F-test for Overall Model Significance

Test $H_0 : \beta_1 = 0$ vs $H_a : \beta_1 \neq 0$ at significant level α

$$F = \frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)} \sim F(1, n-2) \quad \text{under } H_0$$

Reject H_0 if $F > F_\alpha(1, n-2)$ or p-value $< \alpha$.

F Table for alpha=.05



df2/df1	1	2	3	4	5	6	7	8	9	10	12	15	20	24
1	161.4476	199.5000	215.7073	224.5832	230.1619	233.9860	236.7684	238.8827	240.5433	241.8817	243.9060	245.9499	248.0131	249.0518
2	18.5128	19.0000	19.1643	19.2468	19.2964	19.3295	19.3532	19.3710	19.3848	19.3959	19.4125	19.4291	19.4458	19.4541
3	10.1280	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123	8.7855	8.7446	8.7029	8.6602	8.6385
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.0410	5.9988	5.9644	5.9117	5.8578	5.8025	5.7744
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725	4.7351	4.6777	4.6188	4.5581	4.5272
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.0990	4.0600	3.9999	3.9381	3.8742	3.8415
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.8660	3.7870	3.7257	3.6767	3.6365	3.5747	3.5107	3.4445	3.4105
8	5.3177	4.4590	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881	3.3472	3.2839	3.2184	3.1503	3.1152
9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789	3.1373	3.0729	3.0061	2.9365	2.9005
10	4.9646	4.1028	3.7083	3.4780	3.3258	3.2172	3.1355	3.0717	3.0204	2.9782	2.9130	2.8450	2.7740	2.7372
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.9480	2.8962	2.8536	2.7876	2.7186	2.6464	2.6090
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964	2.7534	2.6866	2.6169	2.5436	2.5055
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144	2.6710	2.6037	2.5331	2.4589	2.4202
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458	2.6022	2.5342	2.4630	2.3879	2.3487
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876	2.5437	2.4753	2.4034	2.3275	2.2878
16	4.4940	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377	2.4935	2.4247	2.3522	2.2756	2.2354
17	4.4513	3.5915	3.1968	2.9647	2.8100	2.6987	2.6143	2.5480	2.4943	2.4499	2.3807	2.3077	2.2304	2.1898
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117	2.3421	2.2686	2.1906	2.1497
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779	2.3080	2.2341	2.1555	2.1141
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.5990	2.5140	2.4471	2.3928	2.3479	2.2776	2.2033	2.1242	2.0825
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.3660	2.3210	2.2504	2.1757	2.0960	2.0540
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419	2.2967	2.2258	2.1508	2.0707	2.0283
23	4.2793	3.4221	3.0280	2.7955	2.6400	2.5277	2.4422	2.3748	2.3201	2.2747	2.2036	2.1282	2.0476	2.0050
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002	2.2547	2.1834	2.1077	2.0267	1.9838
25	4.2417	3.3852	2.9912	2.7587	2.6030	2.4904	2.4047	2.3371	2.2821	2.2365	2.1649	2.0889	2.0075	1.9643
26	4.2252	3.3690	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655	2.2197	2.1479	2.0716	1.9898	1.9464
27	4.2100	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501	2.2043	2.1323	2.0558	1.9736	1.9299
28	4.1960	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.2360	2.1900	2.1179	2.0411	1.9586	1.9147
29	4.1830	3.3277	2.9340	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229	2.1768	2.1045	2.0275	1.9446	1.9005
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646	2.0921	2.0148	1.9317	1.8874
40	4.0847	3.2317	2.8387	2.6060	2.4495	2.3359	2.2490	2.1802	2.1240	2.0772	2.0035	1.9245	1.8389	1.7929
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.0970	2.0401	1.9926	1.9174	1.8364	1.7480	1.7001
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.1750	2.0868	2.0164	1.9588	1.9105	1.8337	1.7505	1.6587	1.6084
inf	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799	1.8307	1.7522	1.6664	1.5705	

ANOVA Table

Complete ANOVA Table (10 students data):

Source	df	Sum of Squares	Mean Square	F
Regression	1	631.773	631.773	44.22
Error	8	114.327	14.291	
Total	9	746.100		

F-test:

$$F = \frac{MSR}{MSE} = \frac{631.773}{14.291} = 44.22$$

$$F_{0.05}(1, 8) = 5.32 \quad \text{(from F-table)}$$

$$44.22 > 5.32 \Rightarrow \text{Reject } H_0$$

Conclusion: Significant linear relationship between study hours and exam scores.

Coefficient of Determination (r^2)

The coefficient of determination, or R-squared (r^2), is a statistical measure in a regression model that indicates the proportion or percentage of the variance in the dependent variable that is explained by the independent variable(s).

$$r^2 = \frac{SSR}{TSS} = 1 - \frac{SSE}{TSS} = \left(\frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}\right)^2 = (\text{ Correlation Coefficient})^2. \text{ (See Ch\#3 STAT 2507)}$$

- $0 \leq r^2 \leq 1$: Always between 0 and 1 (or 0% and 100%).
  - 0 (or 0%): Model explains none of the variability.
  - 1 (or 100%): Model explains all the variability (perfect fit).
- A higher r^2 means more variability is explained, but a good value depends on the field: A "good" r^2 in one field (like physics) may be close to 1, while in others (like sociology or psychology) a value of 0.3 might be considered meaningful.
- An $r^2 =$ of 0.72 means 72% of the variation in the dependent variable is predictable from the independent variable.

10 students data:

$$r^2 = \frac{SSR}{TSS} = \frac{631.773}{746.100} = 0.8467 \quad (84.67\%)$$

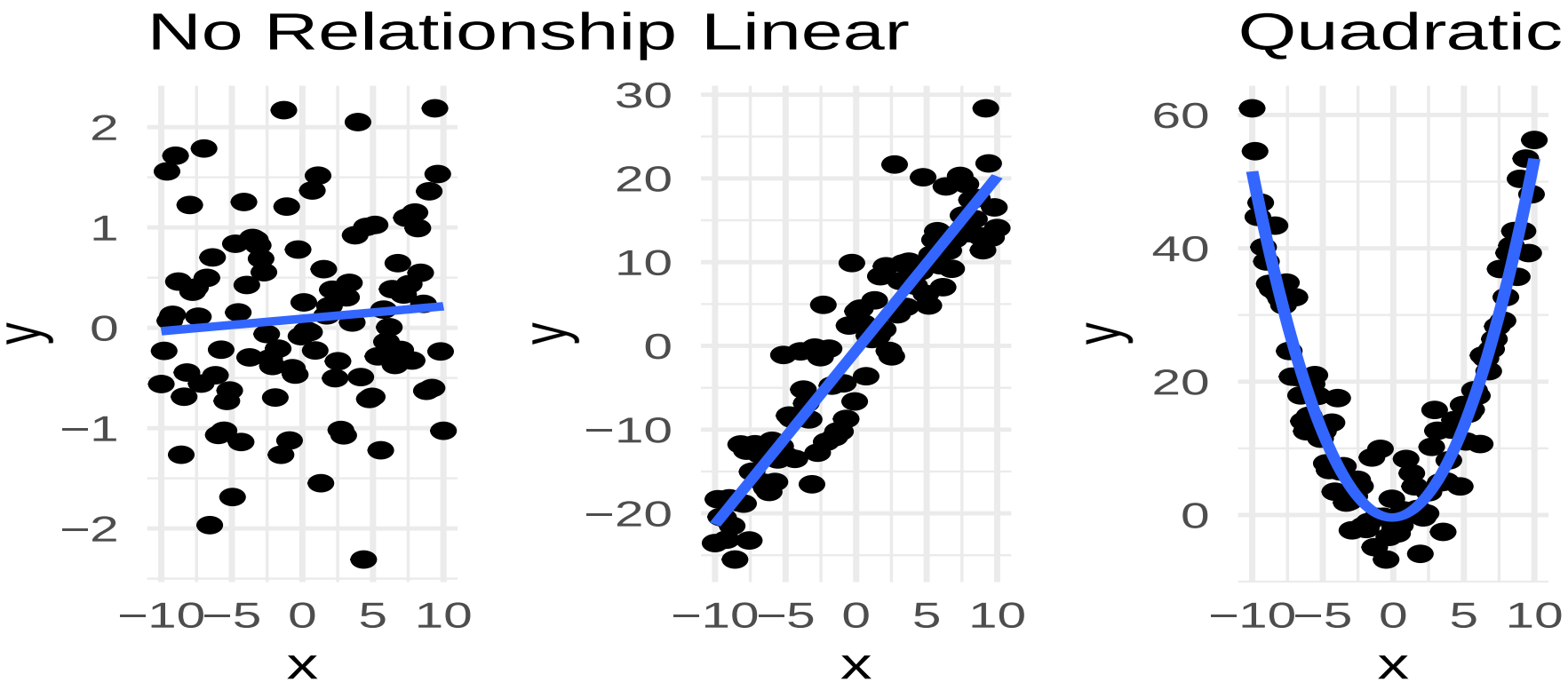
Interpretation: 84.7% of the variation in exam scores is explained by study hours.

Residual Analysis for Checking Model Assumptions

- **Crucial foundation:** All previous statistical results depend on the Simple Linear Regression (SLR) model assumptions being satisfied.
- **Key insight:** If the SLR assumptions are violated, the model won't fit the data properly, and all our conclusions become unreliable.
- **Required diagnostic step:** After fitting the model $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, we must check these assumptions through residual analysis.
- **Binary decision rule:**
  - If violations exist: Discard the fitted model
  - If assumptions hold: Proceed with model interpretation
- **What we expect when assumptions are valid:**
  - Parameter estimates approximate true values: $\hat{\beta}_0 \approx \beta_0$ and $\hat{\beta}_1 \approx \beta_1$,
  - Residuals $e_i = y_i - \hat{y}_i$ mimic the error terms ε_i,
  - Specifically: $e_i \sim N(0, \sigma^2)$ (normal distribution with zero mean, constant variance).

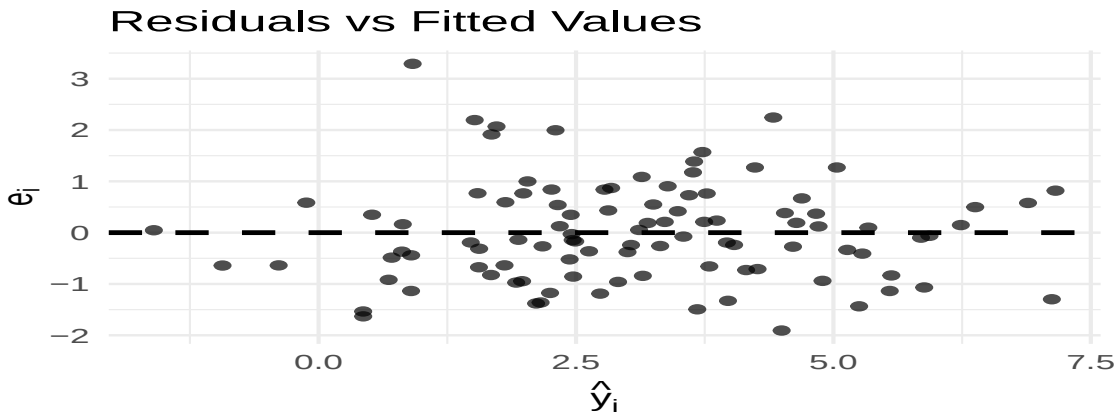
Residual Analysis for Checking Model Assumptions

Note that, before fitting a SLR model, you should always draw a scatter plot of y (vertical) v.s. x (horizontal) to check if y and x have an approximate linear relationship. If yes, you can fit the SLR model and use residual analysis to check Assumptions. Otherwise, you should not fit a SLR model at all.

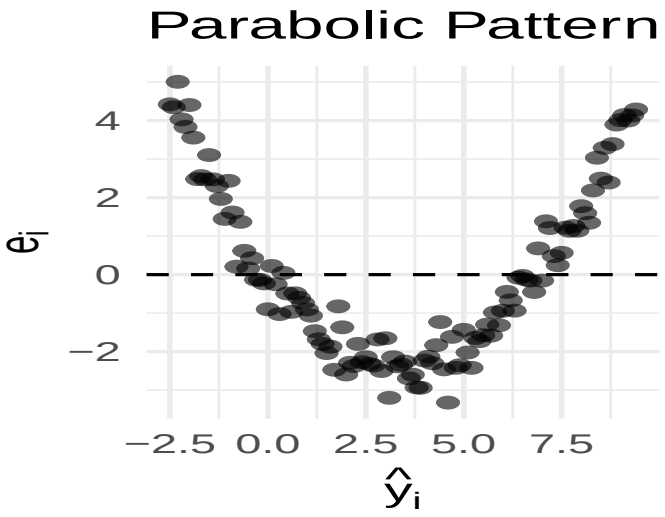


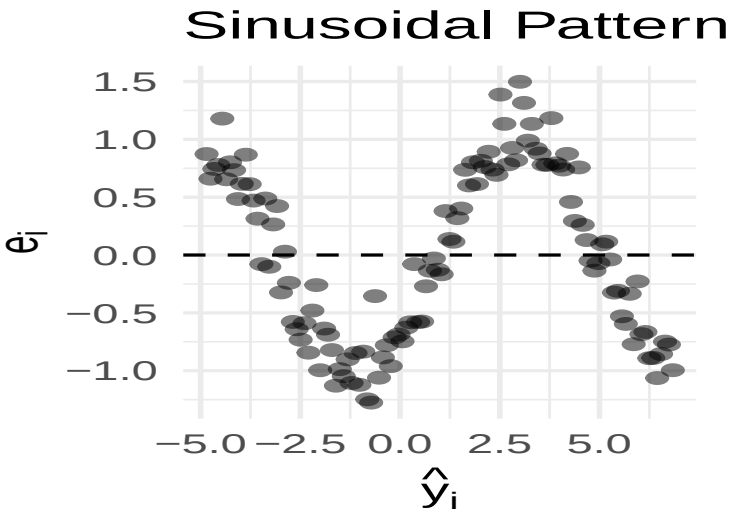
Scatter Plot of e_i's v.s. Fitted Values $\hat{y}_i$

- If SLR model assumption holds, in the plot of e_i (vertical) v.s. fitted value $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ (horizontal), should approximately evenly scattered around without showing any obvious pattern.



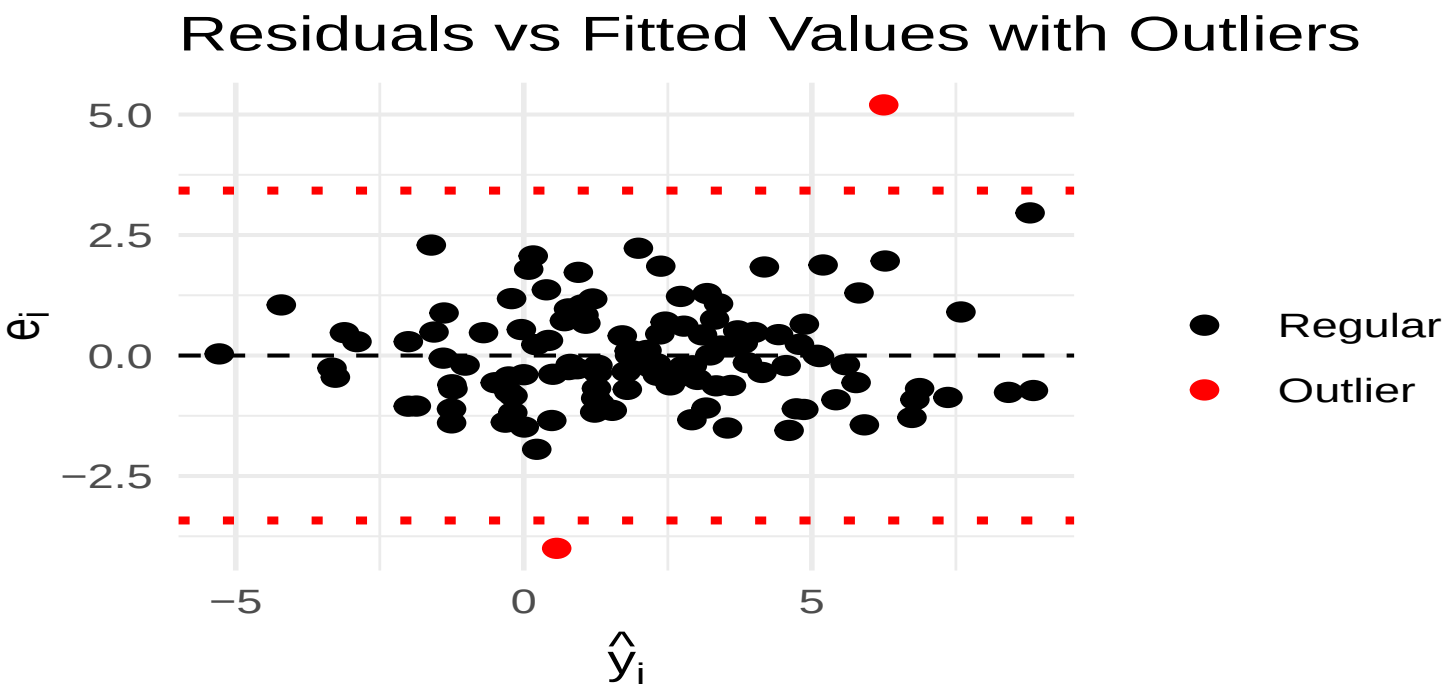
- If the plot of e_i v.s $\hat{y}_i$ shows an obvious pattern, it means that the linear relationship assumption between y and x is not appropriate.





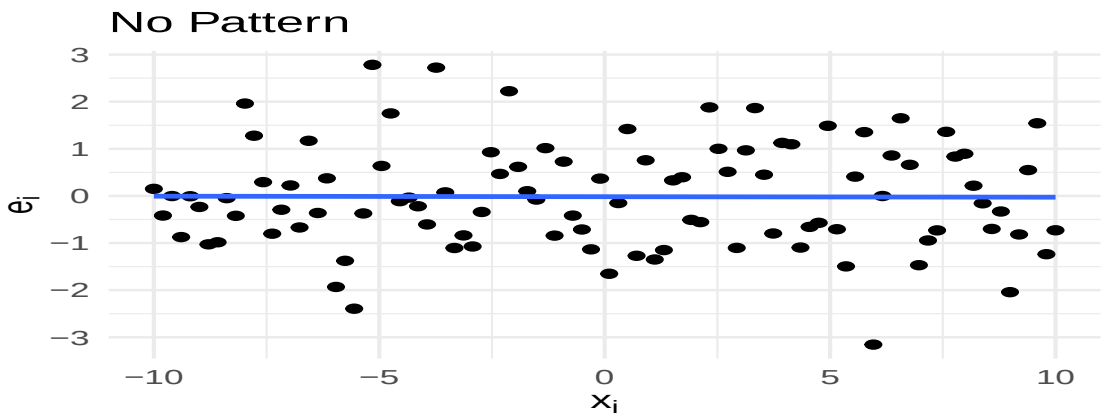
Scatter Plot of e_i's v.s. Fitted Values $\hat{y}_i$

- It can find outliers in the data. Outliers are observations whose residuals lie more than 3 standard deviation from the mean ($mean = 0$).
  - An outlier does not necessarily suggested a violation of SLR model assumption, but it may affect the estimation and inference under SLR model significantly. That is a SLR model fitted with outliers included and one fitted with outliers removed may be very different.
  - When outliers are found, one should be cautious and reporting results one should report the existence of outliers.

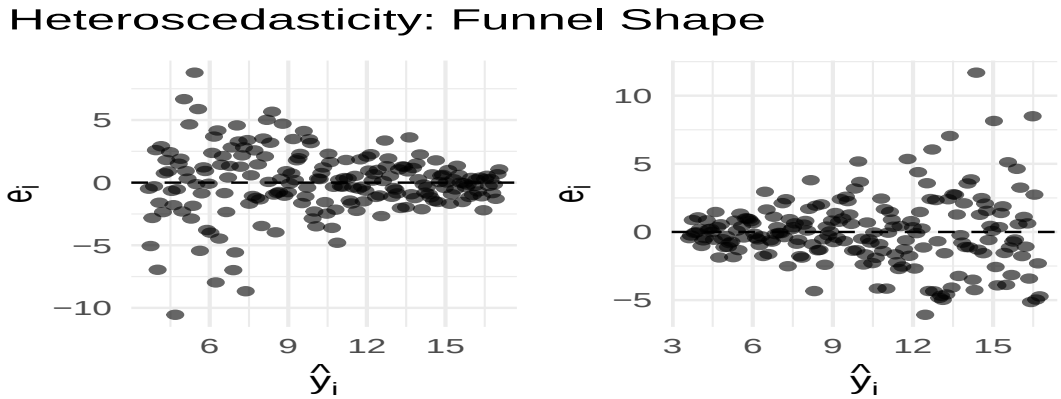


Scatter Plot of e_i's V.s. x_i's

- **Purpose:** Detecting possible violation of constant variance assumption.
- If the SLR model assumptions hold, the scatter plot of e_i's V.s. x_i's should not show any obvious pattern and e_i's should approximately evenly scattered around 0 (which suggests that ε's have constant variance).



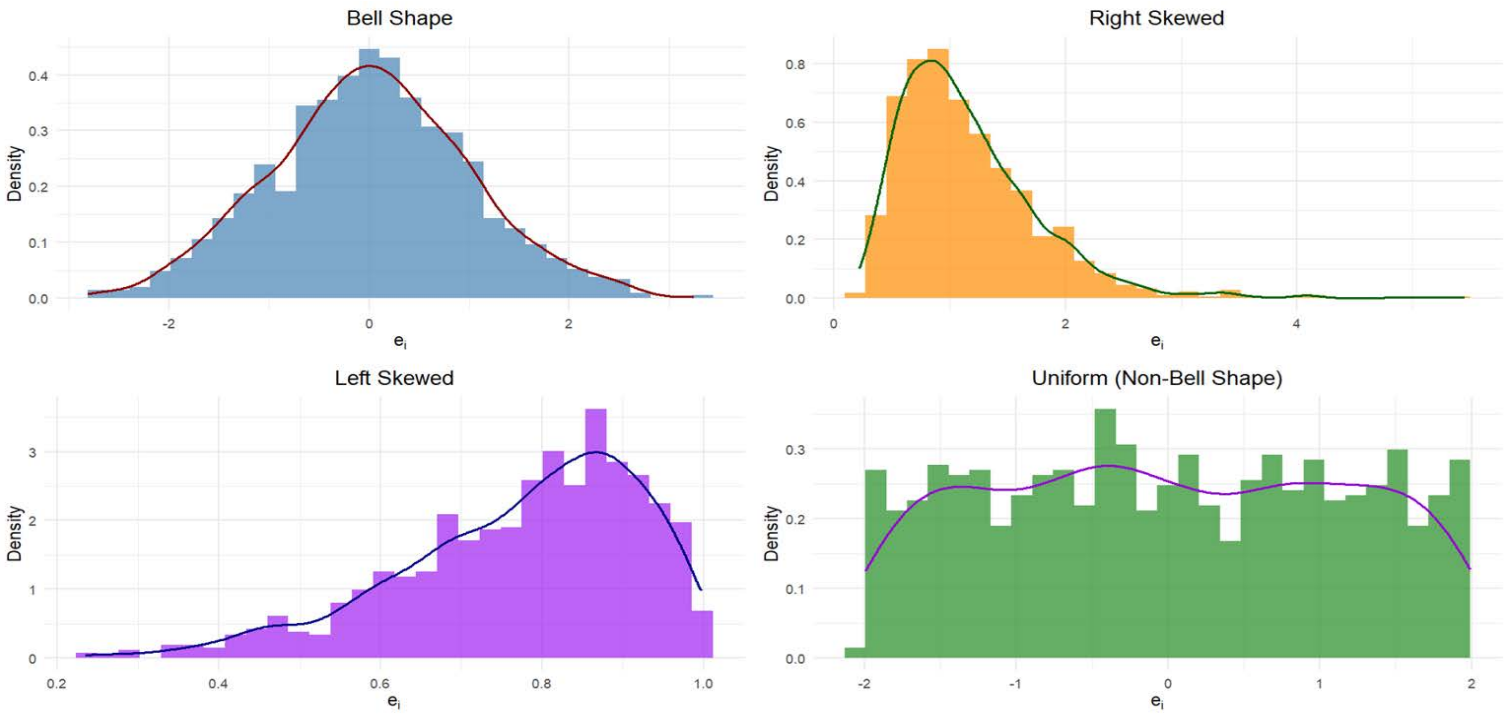
- If the scatter plot of e_i v.s. x_i shows an obvious pattern, it means that the constant variance assumption for ε is violated (Heteroscedasticity problem).



Histogram of ε

Goal: Detecting possible violation of normality assumption.

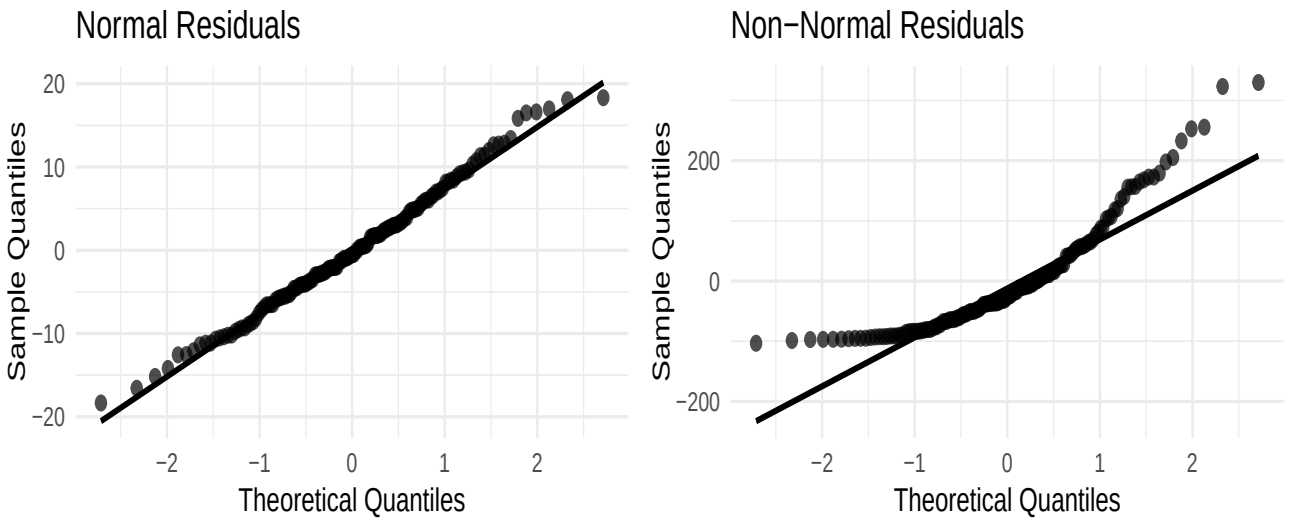
- If SLR model assumptions hold, the histogram of ε should look like the histogram of normal observations, i.e. approximately bell-shaped and symmetric.
- A very non-bell-shaped or non-symmetric histogram of ε indicates that the normality assumption for ε is violated.



Q-Q plot of Residuals

Goal: Detecting possible violation of normality assumption.

- Plot the (sample) quantiles of residuals against theoretical quantiles of the normal distribution.
- If the points on the Q-Q plot falls approximately along a straight line with angle 45, called "reference line", then no obvious violation of normality assumption for ε is found.
- If the points on the Q-Q plot do not approximately fall on the reference line, then it indicates violation of normality assumption for ε.



Possible Remedies for Assumption Violations

- Violation of normality assumption and/or violation of constant (equal) variance assumption might be solved by transforming the response variable y using a suitable function, e.g. $\ln y$, $\frac{1}{y}$, or $\sqrt{y}$, and treat the transformed y as the response variable in the fitting of SLR model.
- Violation of linear relationship assumption might be solved by adding an high-order term. e.g. x^2, as an extra explanatory variable and fit a multiple linear regression model (to be covered in Topic 2).

Lack of Fit Test when x_i's Have Repeated Values

Suppose that the explanatory variable x_i's have repeated values (i.e. we have more than one observations for at least one x value).

Example:

$$\left.\begin{array}{ll} x_1 = 20, & y_1 = 86 \\ x_2 = 40, & y_2 = 78 \\ x_3 = 40, & y_3 = 84 \\ x_4 = 20, & y_4 = 33 \\ x_5 = 60, & y_5 = 64 \end{array}\right\}$$

Two observations, $i = 1$ & 4, have the same x value 20, and two other observations, $i = 2$ & 3 have the same x value 40.

Lack of Fit Test

In this case, we can use a statistical test called **Lack-of-fit** test to test the following hypothesis:

$$H_0 : \text{The SLR model is appropriate}$$
$$Ha : \text{The SLR model is not appropriate}$$

<u>Meanings of H_0 & H_a:</u>

(H_0) means that the SLR model itself and Regression Assumptions all hold true. Thus, the "SLR is appropriate".
(H_a) means that the SLR model or one of Regression Assumptions are violated. Thus, the "SLR is not appropriate".

Setup for the lack-of-fit test

Suppose that there are r $(< n)$ distinct values in $x_1, \cdots, x_n$, and for the i-th, $i = 1, \ldots, r$, distinct value of x, there are n_i observations.
Note that

$$\sum_{i=1}^{r} n_i = n_1 + \cdots + n_r = n.$$

Moreover, for the i-th distinct value of x, denote the corresponding y values as y_{ij}, $j = 1, \ldots, n_i$, and denote

$$\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}.$$

For example, in the previous example, number of distinct values in $x_1, \cdots, x_n$ is $r = 3$. Denote the distinct values of x as x_i, $i = 1, 2, 3$. Then we have

i	x_i	n_i	y_{ij}	$\bar{y}_i$
1	20	2	$86, 33$	$59.5 \left(= \frac{1}{2}(86 + 33)\right)$
2	40	2	$78, 84$	$81 \left(= \frac{1}{2}(78 + 84)\right)$
3	60	1	64	$64 \left(= \frac{64}{1}\right)$

We decompose the SSE of the SLR model into two parts:

- Sum of squares due to pure experimental error ($SSPE$):

$$SSPE := \sum_{i=1}^{r} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

  with corresponding d.f. $\sum_{i=1}^{r}(n_i - 1) = \sum_{i=1}^{r} n_i - r = n - r$.

- Sum of squares due to Lack of fit ($SSLF$):

$$SSLF := SSE - SSPE$$

  with corresponding d.f. $\underbrace{(n-2)}_{\text{d.f. of SSE}} - \underbrace{(n-r)}_{\text{d.f. of SSPE}} = r - 2$.

Example Continued

In the previous example, we have

$$
\begin{aligned}
SSPE &= \sum_{i=1}^{r=3}\sum_{j=1}^{n_i}(y_{ij}-\bar{y}_i)^2 \\
&= (86-59.5)^2+(33-59.5)^2+ && (\leftarrow i=1)\\
&\quad (78-81)^2+(84-81)^2+ && (\leftarrow i=2)\\
&\quad (64-64)^2 && (\leftarrow i=3)\\
&\approx 1422.5.
\end{aligned}
$$

$$S_{yy} = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}(\sum_{i=1}^{n} y_i)^2 = 25721 - \frac{(345)^2}{5} = 1916$$

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{1}{n}(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i) = 12700 - \frac{1}{5}(180)(345) = 280.$$

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2 = 7600 - \frac{1}{5}(180)^2 = 1120$$

$$\Rightarrow SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 1916 - \frac{(280)^2}{1120} = 1846$$

$$SSLF = SSE - SSPE \approx 1846 - 1422.5 = 423.5$$

F-Test for Lack of Fit

Hypotheses

- H_0: Linear model is adequate (no lack of fit)
- H_a: Linear model is inadequate (significant lack of fit)

Test statistic

$$F = \frac{\text{MSLF}}{\text{MSPE}} = \frac{\text{SSLF}/(r-2)}{\text{SSPE}/(n-r)}$$

where:

$$\text{MSLF} = \text{Mean Square Lack-of-Fit} = \frac{\text{SSLF}}{r-2}$$

$$\text{MSPE} = \text{Mean Square Pure Error} = \frac{\text{SSPE}}{n-r}$$

Decision rule

Reject H_0 if:

$$F > F_\alpha(r-2, n-r)$$

where $F_\alpha(r-2, n-r)$ is the critical value from F-distribution. i.e., large $F \Rightarrow$ Significant lack of fit $\Rightarrow$ Linear model may be inadequate.

Remark: The lack-of-fit test can be used only if there are repeated values in $x_1, \ldots, x_n$.

Example Continued

We already calculated: $(r = 3, n = 5)$

$$\text{SSLF} \approx 423.5, \quad \text{d.f.1} = r - 2 = 3 - 2 = 1$$
$$\text{SSPE} \approx 1422.5, \quad \text{d.f.2} = n - r = 5 - 3 = 2.$$
$$\Rightarrow F = \frac{\text{SSLF/d.f.1}}{\text{SSPE/d.f.2}} = \frac{423.5/1}{1422.5/2} \approx 0.5954$$

Critical value when $\alpha = 0.05$ is $F_{0.05;1,2} = 18.51$.
Since $F = 0.5954 < F_{0.05;1,2} = 18.51$, We do fail to reject H_0 at the 0.05 significance level and conclude that we did not find evidence to indicate that the SLR model is not appropriate.

Steps in an SLR Analysis

1. Draw a scatterplot of y against x. If y and x show an approximately linear pattern, proceeds to step (2).
2. State the SLR model $y = \beta_0 + \beta_1 x + \varepsilon$ along with all assumptions required for inference under SLR.
3. Use the observed data to find the least-squares fitted line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

4. Perform a residual analysis to check for violations of assumptions. If no violations found, go to step (5). Otherwise, apply suitable remedies and go back to step (2) to start over.
5. Test usefulness of model for predictive purposes:
$H_0 : \beta_1 = 0$ VS $H_a : \beta_1 \neq 0$, and calculate

$$r^2 = \frac{\text{SSR}}{\text{TSS}}.$$

6. If $H_0 : \beta_1 = 0$ is rejected in step (5), use the SLR model for prediction.

Simple Linear Regression (SLR) Formulae

- Intercept estimator:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- Slope estimator:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

- Error variance estimator:

$$s^2 = \frac{SSE}{n-2} \quad \text{where} \quad SSE = S_{yy} - \frac{(S_{xy})^2}{S_{xx}}$$

$$S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n}, \quad S_{yy} = \sum y_i^2 - \frac{(\sum y_i)^2}{n}, \quad S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$$

- Test statistic for β_1:

$$t = \frac{\hat{\beta}_1}{s/\sqrt{S_{xx}}} \sim t_{(n-2)}$$

- $100(1-\alpha)\%$ confidence interval for β_1:

$$\hat{\beta}_1 \pm t_{\alpha/2,\, n-2} \cdot \frac{s}{\sqrt{S_{xx}}}$$

- Total, regression, and error sums of squares:

$$TSS = S_{yy}, \quad SSR = \frac{(S_{xy})^2}{S_{xx}}, \quad F = \frac{MSR}{MSE} \sim F_{(1,n-2)}$$

- $100(1-\alpha)\%$ confidence interval for mean response $E(y \mid x^\star)$:

$$\hat{y} \pm t_{\alpha/2,\, n-2}\sqrt{MSE\left(\frac{1}{n}+\frac{(x^\star-\bar{x})^2}{S_{xx}}\right)}$$

- $100(1-\alpha)\%$ prediction interval for individual $y \mid x^\star$:

$$\hat{y} \pm t_{\alpha/2,\, n-2}\sqrt{MSE\left(1+\frac{1}{n}+\frac{(x^\star-\bar{x})^2}{S_{xx}}\right)}$$

- Coefficient of correlation:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

- Coefficient of determination:

$$r^2 = \frac{SSR}{TSS}$$

- Lack-of-fit decomposition:

$$SSE = SSPE + SSLF, \qquad \text{where } SSPE = \sum_i \sum_j (y_{ij}-\bar{y}_i)^2$$

- Lack-of-fit F-test:

$$F_{LF} = \frac{MSSLF}{MSSPE} = \frac{SSLF/\left[n-2-\sum(n_i-1)\right]}{SSPE/\left[\sum(n_i-1)\right]} \sim F_{(n-2-\sum(n_i-1),\, \sum(n_i-1))}$$